

```
PS D:\folder\rac\ih\môn\hk1-4\ptht-phantan\LT\bt\b4\1> java Main
```

```
=====
CHUONG TRINH KIEM TRA HE THONG MESSAGE BROKER (BAI TAP 4)
=====
```

```
>>> 1. KIEM TRA 1.1: CONSUMER DANG CHAY <<<
```

```
[Customer] Sending ORDER-001
```

```
[Broker] Message received ORDER-001
```

```
[OrderService] Received ORDER-001
```

```
[OrderService] Completed ORDER-001
```

```
[Customer] Sending ORDER-002
```

```
[Broker] Message received ORDER-002
```

```
[OrderService] Received ORDER-002
```

```
[OrderService] Completed ORDER-002
```

```
[Customer] Sending ORDER-003
```

```
[Broker] Message received ORDER-003
```

```
[OrderService] Received ORDER-003
```

```
[OrderService] Completed ORDER-003
```

```
-----
>>> 2. KIEM TRA 1.2: CONSUMER CHUA CHAY (MESSAGE QUEUE) <<<
```

```
>>> 2. KIEM TRA 1.2: CONSUMER CHUA CHAY (MESSAGE QUEUE) <<<
```

```
--> Cho Customer gui message truoc (Chua chay OrderService):
```

```
[Customer] Sending ORDER-001
```

```
[Broker] Message received ORDER-001
```

```
[Customer] Sending ORDER-002
```

```
[Broker] Message received ORDER-002
```

```
[Customer] Sending ORDER-003
```

```
[Broker] Message received ORDER-003
```

```
--> Sau do moi khoi dong OrderService:
```

```
Ket qua:
```

```
[OrderService] Received ORDER-001
```

```
[OrderService] Received ORDER-002
```

```
[OrderService] Received ORDER-003
```

```
PS D:\folder\rac\iuu\mon\hk1-4\ptht-phantan\LT\bt\b4\1> java Test3
=== KIEM TRA 3: NHIEU NGUOI TIEU DUNG (COMPETING CONSUMERS) ===
Mo hinh: 1 Message Broker phan phoi cho 2 Service (Service-1 va Service-2)

[Customer] Sending ORDER-001
[Broker] Message received ORDER-001
[Service-1] Received ORDER-001
[Service-1] Completed ORDER-001
ORDER-001 -> Service-1

[Customer] Sending ORDER-002
[Broker] Message received ORDER-002
[Service-2] Received ORDER-002
[Service-2] Completed ORDER-002
ORDER-002 -> Service-2

[Customer] Sending ORDER-003
[Broker] Message received ORDER-003
[Service-1] Received ORDER-003
[Service-1] Completed ORDER-003
ORDER-003 -> Service-1
```

```
OrderService.java U Main.java U Customer.java U X
h > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > Customer.java
1 public class Customer {
2     private MessageBroker broker;
3
4     public Customer(MessageBroker broker) {
5         this.broker = broker;
6     }
7
8     /**
9     */
10    public void sendOrder(String orderId) {
11        System.out.println("[Customer] Sending " + orderId);
12        broker.sendMessage(orderId);
13    }
14 }
15
```

```
Customer.java U MessageBroker.java U InventoryService.java U NotificationService.java U
iuh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > InventoryService.java
1 public class InventoryService implements OrderSubscriber {
2     @Override
3     public void onOrderReceived(Order order) {
4         System.out.println("[InventoryService]");
5         System.out.println("Received " + order.getId());
6         System.out.println("Checking and updating inventory for product: " + order.getId());
7         try {
8             Thread.sleep(700); // Mo phong thoi gian kiem tra va cap nhap kho
9         } catch (InterruptedException e) {
10            Thread.currentThread().interrupt();
11        }
12        System.out.println("Inventory updated.");
13        System.out.println();
14    }
15 }
16
```

```
rac - Antigravity IDE - Main.java
Main.java M X README.md M
iuh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > Main.java
1 public class Main {
2     public static void main(String[] args) {
3         System.out.println("=====");
4         System.out.println("CHƯƠNG TRÌNH KIỂM TRA HỆ THỐNG MESSAGE BROKER (BAI TAP 4)");
5         System.out.println("=====");
6
7         // -----
8         // KIỂM TRA 1.1: CONSUMER ĐANG CHẠY
9         // -----
10        System.out.println(">>> 1. KIỂM TRA 1.1: CONSUMER ĐANG CHẠY <<<");
11        MessageBroker broker1 = new MessageBroker();
12        OrderService orderService1 = new OrderService("OrderService");
13        broker1.registerConsumer(orderService1);
14
15        Customer customer1 = new Customer(broker1);
16        customer1.sendOrder("ORDER-001");
17        customer1.sendOrder("ORDER-002");
18        customer1.sendOrder("ORDER-003");
19
20        System.out.println("\n-----\n");
21
22        // -----
23        // KIỂM TRA 1.2: CONSUMER CHƯA CHẠY (TIME-UNCOUPLED)
24        // -----
25        System.out.println(">>> 2. KIỂM TRA 1.2: CONSUMER CHƯA CHẠY (MESSAGE QUEUE) <<<");
26        MessageBroker broker2 = new MessageBroker();
27        Customer customer2 = new Customer(broker2);
28
29        System.out.println(">>> Cho Customer gửi message trước (Chưa chạy OrderService):");
30        customer2.sendOrder("ORDER-001");
31        customer2.sendOrder("ORDER-002");
32        customer2.sendOrder("ORDER-003");
33
34        System.out.println("\n--> Sau đó mới khởi động OrderService:");
35        System.out.println("Ket qua:");
36        OrderService orderService2 = new OrderService("OrderService");
37        broker2.registerConsumer(orderService2);
38
39        System.out.println("\n-----\n");
40    }
41 }
```

```
File Edit Selection View Go Run Terminal Help
Main.java M MessageBroker.java M x README.md M
juh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > MessageBroker.java
1 import java.util.ArrayList;
2 import java.util.LinkedList;
3 import java.util.List;
4 import java.util.Queue;
5
6 public class MessageBroker {
7     private final Queue<String> messageQueue = new LinkedList<>();
8     private final List<OrderService> consumers = new ArrayList<>();
9     private int currentConsumerIndex = 0;
10
11     public void sendMessage(String orderId) {
12         System.out.println("[Broker] Message received " + orderId);
13
14         if (!consumers.isEmpty()) {
15             // Phan phoi luan phien Round-Robin (Moi order chi do 1 Consumer xu ly)
16             OrderService consumer = consumers.get(currentConsumerIndex);
17             currentConsumerIndex = (currentConsumerIndex + 1) % consumers.size();
18             consumer.processOrder(orderId);
19         } else {
20             messageQueue.add(orderId);
21         }
22     }
23
24     public void registerConsumer(OrderService consumer) {
25         this.consumers.add(consumer);
26
27         // Xu ly cac message dang cho san trong hang doi
28         while (!messageQueue.isEmpty()) {
29             String pendingOrder = messageQueue.poll();
30             consumer.receivePendingOrder(pendingOrder);
31         }
32     }
33 }
34
```



```

Main.java M  OrderService.java M X  README.md M
iuh > môn > hk1-4 > ptth-phantan > LT > bt > b4 > 1 > OrderService.java
1  public class OrderService {
2      private String name = "OrderService";
3
4      public OrderService() {
5      }
6
7      public OrderService(String name) {
8          this.name = name;
9      }
10
11     public String getName() {
12         return name;
13     }
14
15     /**
16      * Xu ly don hang khi Consumer dang chay (Kiem tra 1.1 va Kiem tra 3)
17      */
18     public void processOrder(String orderId) {
19         System.out.println "[" + name + "] Received " + orderId;
20         System.out.println "[" + name + "] Completed " + orderId;
21         System.out.println (orderId + " -> " + name);
22         System.out.println();
23     }
24
25     /**
26      * Nhan cac don hang ton dong trong Queue khi vua khoi dong (Kiem tra 1.2)
27      */
28     public void receivePendingOrder(String orderId) {
29         System.out.println "[" + name + "] Received " + orderId;
30     }
31 }
32
```

```

NotificationService.java U  Order.java U  PaymentService.java U  OrderSubscriber.java U X
iuh > môn > hk1-4 > ptth-phantan > LT > bt > b4 > 1 > OrderSubscriber.java
1  public interface OrderSubscriber {
2      /**
3       * Nhan va xu ly don hang duoc phat tu Message Broker
4       * @param order Thông tin don hang
5       */
6      void onOrderReceived(Order order);
7  }
8
```

NotificationService.java U

Order.java U

PaymentService.java U X

OrderSubscriber

ih > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > PaymentService.java

```
1 public class PaymentService implements OrderSubscriber {
2     @Override
3     public void onOrderReceived(Order order) {
4         System.out.println("[PaymentService]");
5         System.out.println("Received " + order.getId());
6         System.out.println("Processing payment...");
7         try {
8             Thread.sleep(800); // Mô phỏng thời gian xử lý thanh toán
9         } catch (InterruptedException e) {
10             Thread.currentThread().interrupt();
11         }
12         System.out.println("Payment completed.");
13         System.out.println();
14     }
15 }
16
```



```
Test1_1.java U X OrderService.java U Main.java U Customer.java U MessageBroker.java U
iuh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > Test1_1.java
1 public class Test1_1 {
2     public static void main(String[] args) {
3         System.out.println("=== KIEM TRA 1.1: CONSUMER DANG CHAY ===");
4
5         MessageBroker broker = new MessageBroker();
6
7         OrderService orderService = new OrderService();
8         broker.registerConsumer(orderService);
9
10        Customer customer = new Customer(broker);
11        customer.sendOrder("ORDER-001");
12        customer.sendOrder("ORDER-002");
13        customer.sendOrder("ORDER-003");
14    }
15 }
16
```

```
Test2_2.java U X Test1_1.java U OrderService.java U Main.java U Customer.java U
iuh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > Test1_2.java
1 public class Test1_2 {
2     public static void main(String[] args) {
3         System.out.println("=== KIEM TRA 1.2: CONSUMER CHUA CHAY ===");
4
5         MessageBroker broker = new MessageBroker();
6         Customer customer = new Customer(broker);
7
8         System.out.println("-> Customer gui 3 don hang vao Broker trong khi OrderService CHUA BAT:");
9         customer.sendOrder("ORDER-001");
10        customer.sendOrder("ORDER-002");
11        customer.sendOrder("ORDER-003");
12
13        System.out.println("\n--> Sau do moi khoi dong OrderService (Lay toan bo message ton trong Queue):");
14        OrderService orderService = new OrderService();
15        broker.registerConsumer(orderService);
16    }
17 }
18
```

```
2 Main.java M Test3.java U X README.md M
uh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > Test3.java

1 public class Test3 {
2     public static void main(String[] args) {
3         System.out.println("=== KIEM TRA 3: NHIEU NGUOI TIEU DUNG (COMPETING CONSUMERS) ===");
4         System.out.println("Mo hinh: 1 Message Broker phan phoi cho 2 Service (Service-1 va Service-2)\n");
5
6         // 1. Khoi tao Message Broker
7         MessageBroker broker = new MessageBroker();
8
9         // 2. Them OrderService-1 va OrderService-2
10        OrderService service1 = new OrderService("Service-1");
11        OrderService service2 = new OrderService("Service-2");
12        broker.registerConsumer(service1);
13        broker.registerConsumer(service2);
14
15        // 3. Customer gui 4 don hang
16        Customer customer = new Customer(broker);
17        customer.sendOrder("ORDER-001");
18        customer.sendOrder("ORDER-002");
19        customer.sendOrder("ORDER-003");
20        customer.sendOrder("ORDER-004");
21
22        System.out.println("--> Ghi chu: Moi don hang chi duoc 1 Service nhan va xu ly (khong bi trung lap).");
23    }
24 }
25
```